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SUBJECT: GENERATIVE AI

COLLEGE: NCRD's Sterling Institute of Management
Studies

ASSIGNMENT NO: 1

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SR NO	TITLE
	<u>CHATBOT</u>
1	Language: Python
2	Tool : Google Colab
3	API-KEY: Gemini 3.5 Flash Lite
4	Libraries : LangChain,Google,Generative AI

PROGRAM DESCRIPTION:

This program uses the Gemini API with Python in Google Colab to send a text prompt and generate an AI-powered response. It authenticates using a Google AI Studio API key and displays the generated content using the Gemini language model.

Chatbot Assignment1.ipynb

CODE:

```
!pip install -U langchain-google-genai
```

```
!pip install -U langchain
```

```
from google.colab import userdata
```

```
from langchain_google_genai import ChatGoogleGenerativeAI
```

1. Setup model with your secret key

```
llm = ChatGoogleGenerativeAI( model="gemini-3.5-flash-  
lite", google_api_key=userdata.get('Adi') )
```

2. Simple list to track context

```
messages = []
```

```
print("Chatbot ready! Type 'exit' to quit.\n")
```

3. Chat loop

```
while True:
```

```
    user_input = input("You: ")
```

```
    if user_input.lower() == "exit":
```

```
        break
```

```
# Add user message as a tuple: ("user", "text")  
messages.append(("user", user_input))
```

```
# Send conversation history to model  
response = llm.invoke(messages)
```

```
# Add AI response to history as a tuple: ("assistant", "text")  
messages.append(("assistant", response.content))
```

```
# Print response  
print(f"Bot: {response.content}\n")
```

OUTPUT:

You: what are the principles of management

WARNING:google_genai.models:Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend

Bot: [{"type": "text", "text": "The **Principles of Management** are broad and general guidelines for managerial decision-making and behavior."}]

You: example of a real world company failing to follow one of these principles

Bot: [{"type": "text", "text": "A classic and dramatic real-world example of a company failing to follow a core management principle is **Boeing**"}]

You:

REFERENCES:

1. References

<https://github.com/Adi-Dev-hub/GenAI>

<https://aistudio.google.com/>

<https://aistudio.google.com/docs>